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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Komali Varnika
Roll Number	2520090010
Branch	CSIT
Course Outcome	C02-Advanced Multi-Way Search Trees (B+ Trees)

Case Study Topic:

B+ Tree Indexing Optimization — Managing Node Split and Page Capacities for High-Throughput Databases

Work Done Summary:

In this case study, we implemented and analyzed a multi-way B+ Tree index tailored for database storage systems. We evaluated its structural write path—focusing on in-place leaf page modifications and structural node splits—and calculated block storage capacities based on pointer and key sizes. We demonstrated how keeping routing keys in internal nodes and actual data records strictly in leaf nodes ensures optimal cache efficiency and fast range query performance.

Implementation Details:

1. Created a multi-purpose node structure to act as either an internal routing branch or a terminal leaf.
2. Built a recursive search function to navigate the tree layers and locate the correct leaf page for insertion.
3. Designed an automatic sorting routine to arrange keys sequentially as they are placed inside a node.
4. Added an automatic splitting mechanism that divides full nodes at the midpoint to maintain balancing.
5. Connected all leaf nodes horizontally using sequential pointers to allow direct data scans without backtracking.

Result: Screenshots/output

```
[5:C] [10:A] [15:D] [20:B]

=== Code Execution Successful ===
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GitHub Link:https://github.com/Varnika1024/DSA_CASE-STUDY/blob/main/co2_assignment.java